Institut Universitaire des sciences

- Faculté des sciences et de technologie

✓ TD N°9 Réseau II

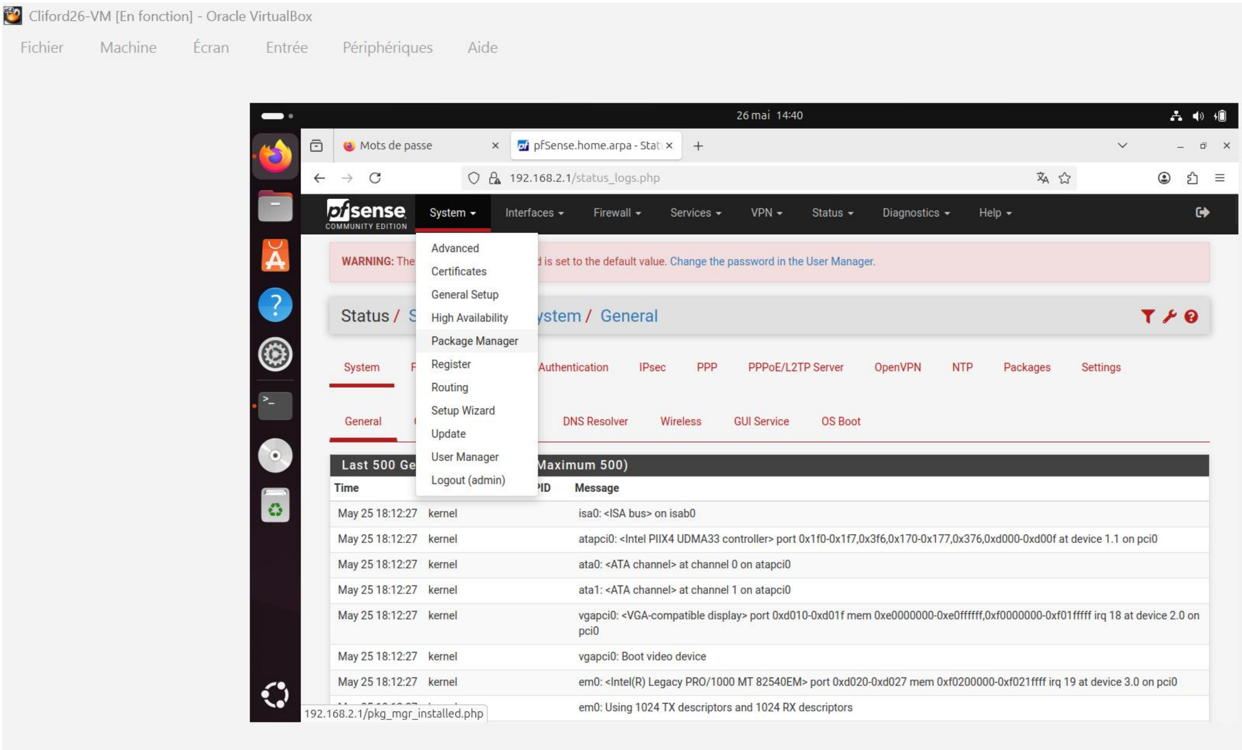
Nom & Prénom : COFFY Cliford

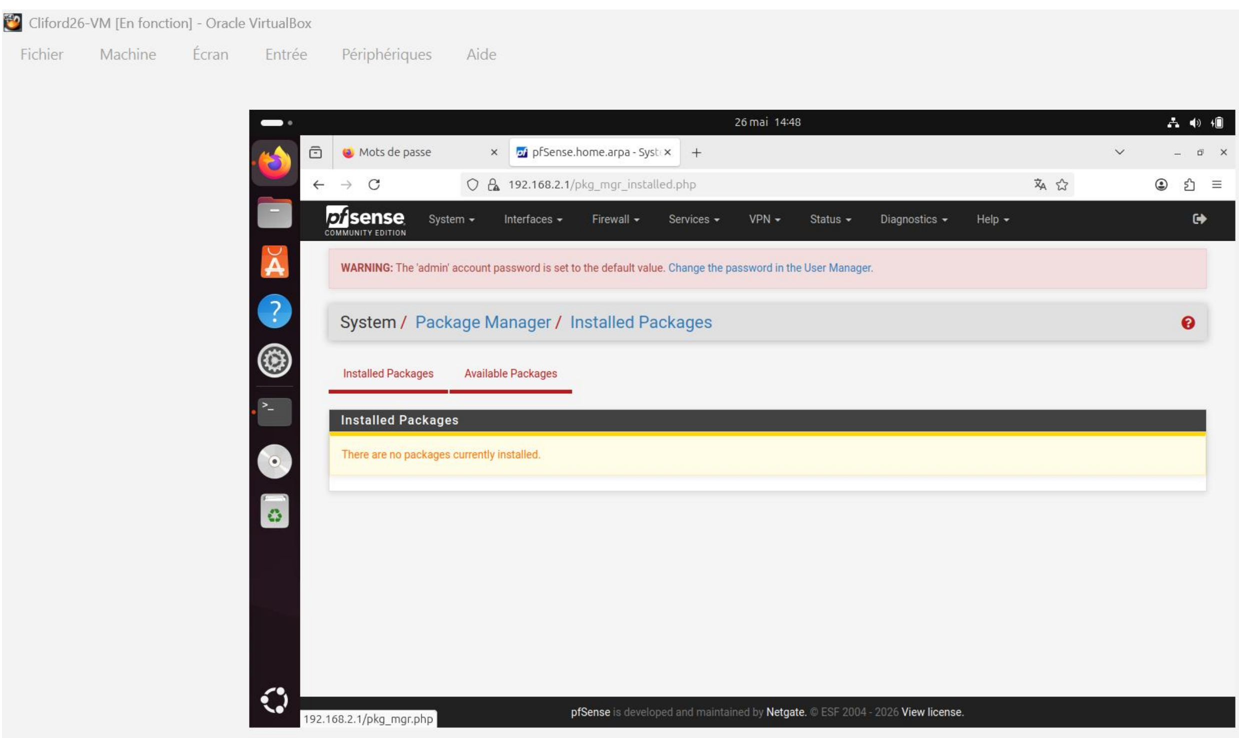
Niveau : L3

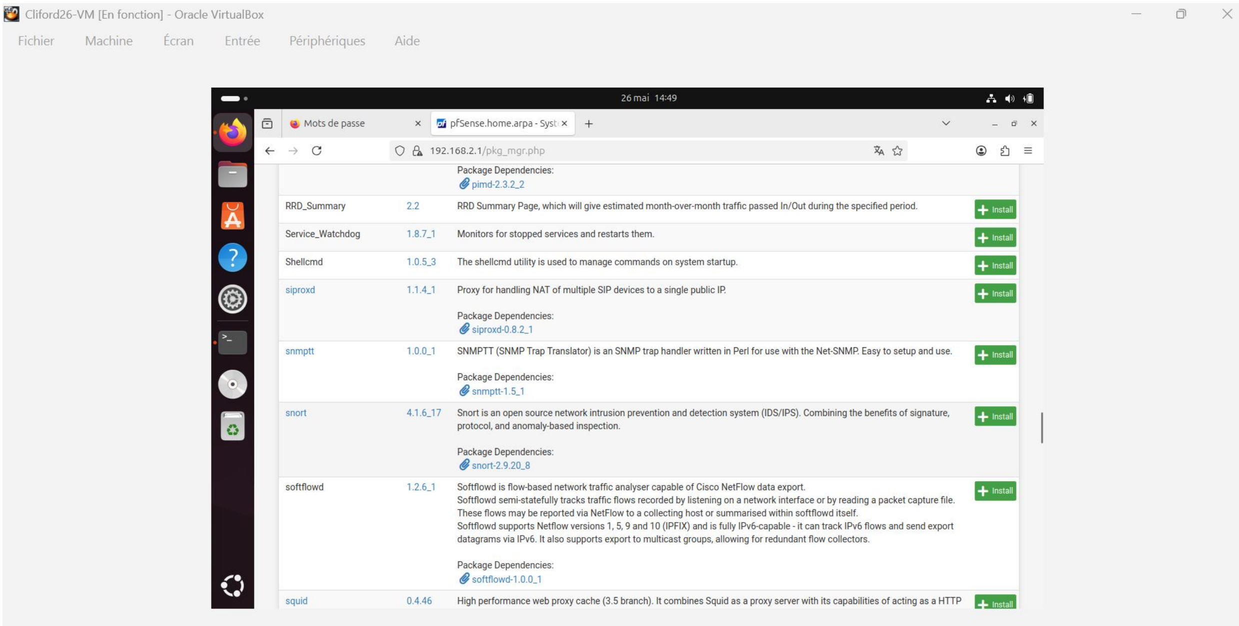
Prof. : Ismaël SAINT AMOUR

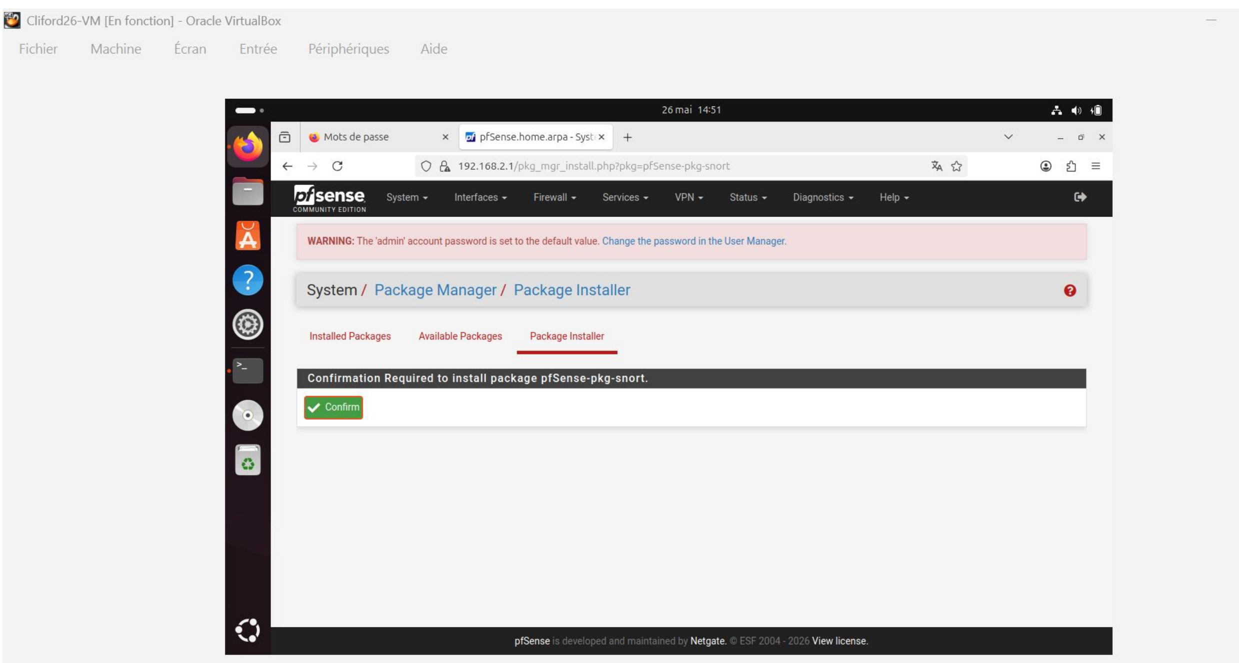
Date : 19/05/2026

Étape 1 : Installation du paquet Snort sur pfSense






Cliford26-VM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
26 mai 14:49
Mots de passe
pfSense.home.arpa - Syst
192.168.2.1/pkg_mgr.php
Package Dependencies:
pimd-2.3.2_2
RRD_Summary 2.2 RRD Summary Page, which will give estimated month-over-month traffic passed In/Out during the specified period.
Install
Service_Watchdog 1.8.7_1 Monitors for stopped services and restarts them.
Shellcmd 1.0.5_3 The shellcmd utility is used to manage commands on system startup.
siproxd 1.1.4_1 Proxy for handling NAT of multiple SIP devices to a single public IP.
Package Dependencies:
siproxd-0.8.2_1
snmptt 1.0.0_1 SNMPTT (SNMP Trap Translator) is an SNMP trap handler written in Perl for use with the Net-SNMP. Easy to setup and use.
Package Dependencies:
snmptt-1.5_1
snort 4.1.6_17 Snort is an open source network intrusion prevention and detection system (IDS/IPS). Combining the benefits of signature, protocol, and anomaly-based inspection.
Package Dependencies:
snort-2.9.20_8
softflowd 1.2.6_1 Softflowd is flow-based network traffic analyser capable of Cisco NetFlow data export.
Softflowd semi-statefully tracks traffic flows recorded by listening on a network interface or by reading a packet capture file.
These flows may be reported via NetFlow to a collecting host or summarised within softflowd itself.
Softflowd supports Netflow versions 1, 5, 9 and 10 (IPFIX) and is fully IPv6-capable - it can track IPv6 flows and send export datagrams via IPv6. It also supports export to multicast groups, allowing for redundant flow collectors.
Package Dependencies:
softflowd-1.0.0_1
squid 0.4.46 High performance web proxy cache (3.5 branch). It combines Squid as a proxy server with its capabilities of acting as a HTTP


Cliford26-VM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
26 mai 14:51
Mots de passe
pfSense.home.arpa - Syst
192.168.2.1/pkg_mgr_install.php?pkg=pfSense-pkg-snort
pfsense COMMUNITY EDITION
System Interfaces Firewall Services VPN Status Diagnostics Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
System / Package Manager / Package Installer
Installed Packages Available Packages Package Installer
Confirmation Required to install package pfSense-pkg-snort.
Confirm
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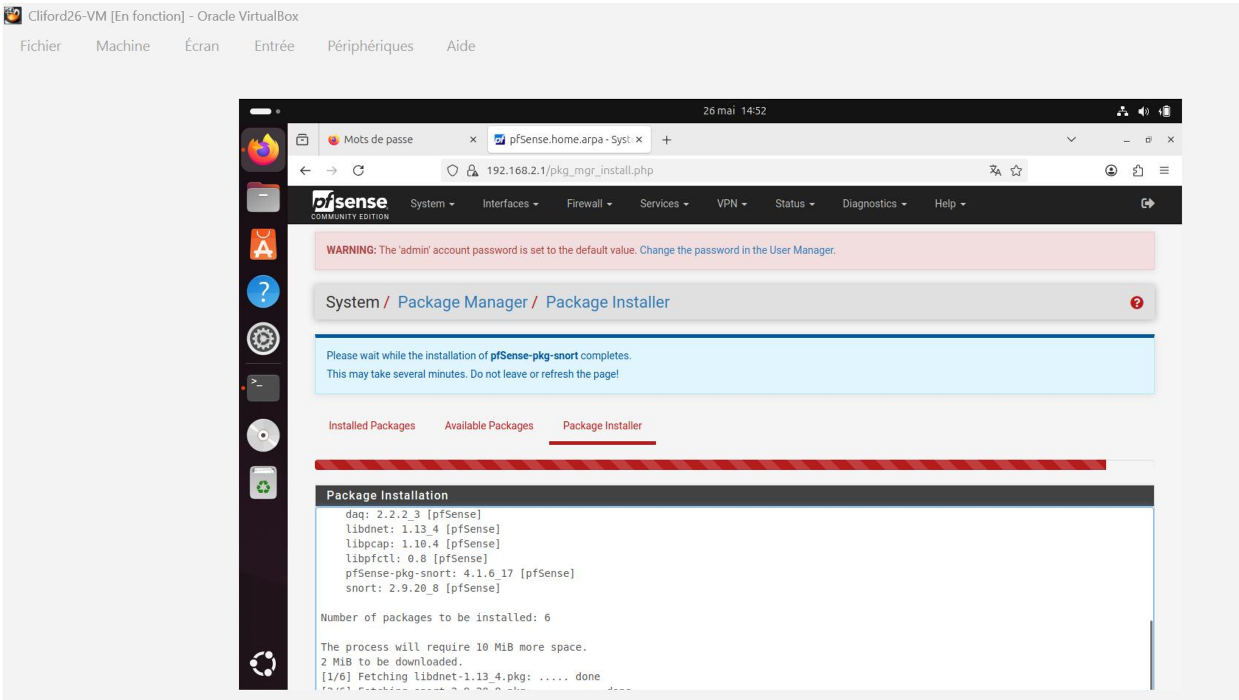
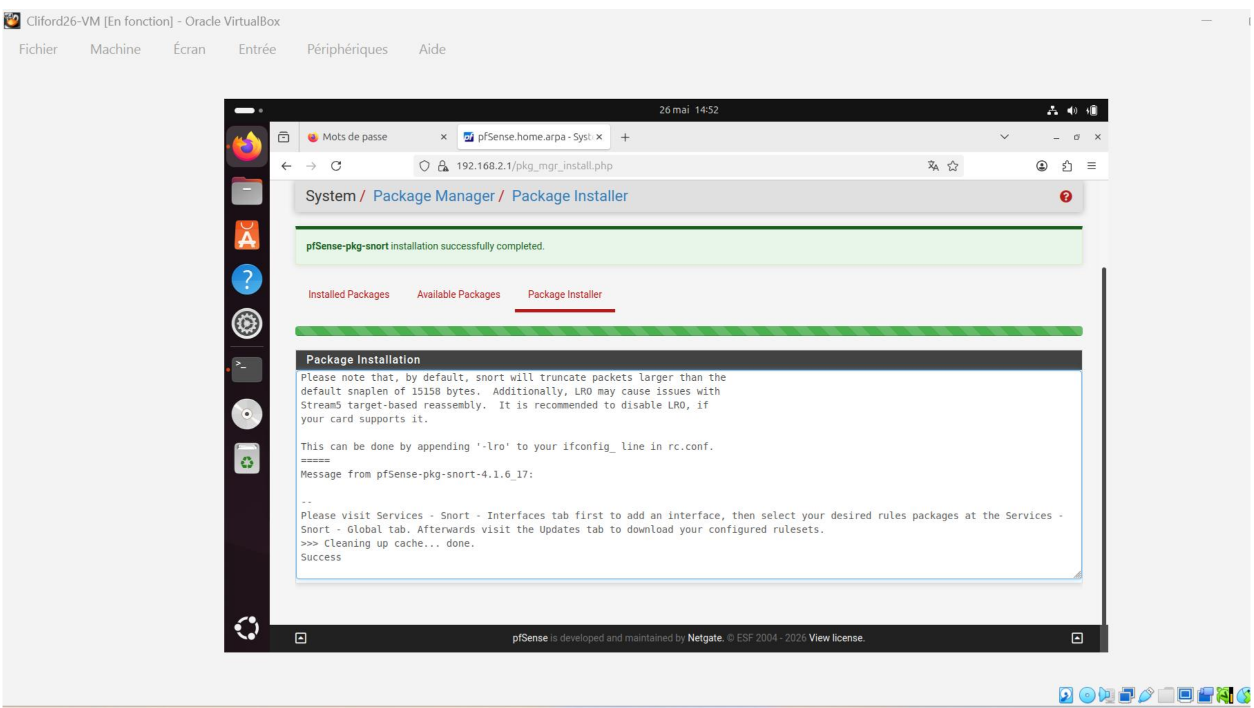

Cliford26-VM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
26 mai 14:52
Mots de passe
pfSense.home.arpa - Syst
192.168.2.1/pkg_mgr_install.php
pfsense COMMUNITY EDITION
System Interfaces Firewall Services VPN Status Diagnostics Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
System / Package Manager / Package Installer
Please wait while the installation of pfSense-pkg-snort completes.
This may take several minutes. Do not leave or refresh the page!
Installed Packages
Available Packages
Package Installer
Package Installation
daq: 2.2.2_3 [pfSense]
libdnet: 1.13_4 [pfSense]
libpcap: 1.10.4 [pfSense]
libpfctl: 0.8 [pfSense]
pfSense-pkg-snort: 4.1.6_17 [pfSense]
snort: 2.9.20_8 [pfSense]
Number of packages to be installed: 6
The process will require 10 MiB more space.
2 MiB to be downloaded.
[1/6] Fetching libdnet-1.13_4.pkg: ..... done


Cliford26-VM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
26 mai 14:52
Mots de passe
pfSense.home.arpa - Syst
192.168.2.1/pkg_mgr_install.php
System / Package Manager / Package Installer
pfSense-pkg-snort installation successfully completed.
Installed Packages
Available Packages
Package Installer
Package Installation
Please note that, by default, snort will truncate packets larger than the
default snaplen of 15158 bytes. Additionally, LRO may cause issues with
Stream5 target-based reassembly. It is recommended to disable LRO, if
your card supports it.
This can be done by appending '-lro' to your ifconfig_ line in rc.conf.
=====
Message from pfSense-pkg-snort-4.1.6_17:
--
Please visit Services - Snort - Interfaces tab first to add an interface, then select your desired rules packages at the Services -
Snort - Global tab. Afterwards visit the Updates tab to download your configured rulesets.
>>> Cleaning up cache... done.
Success
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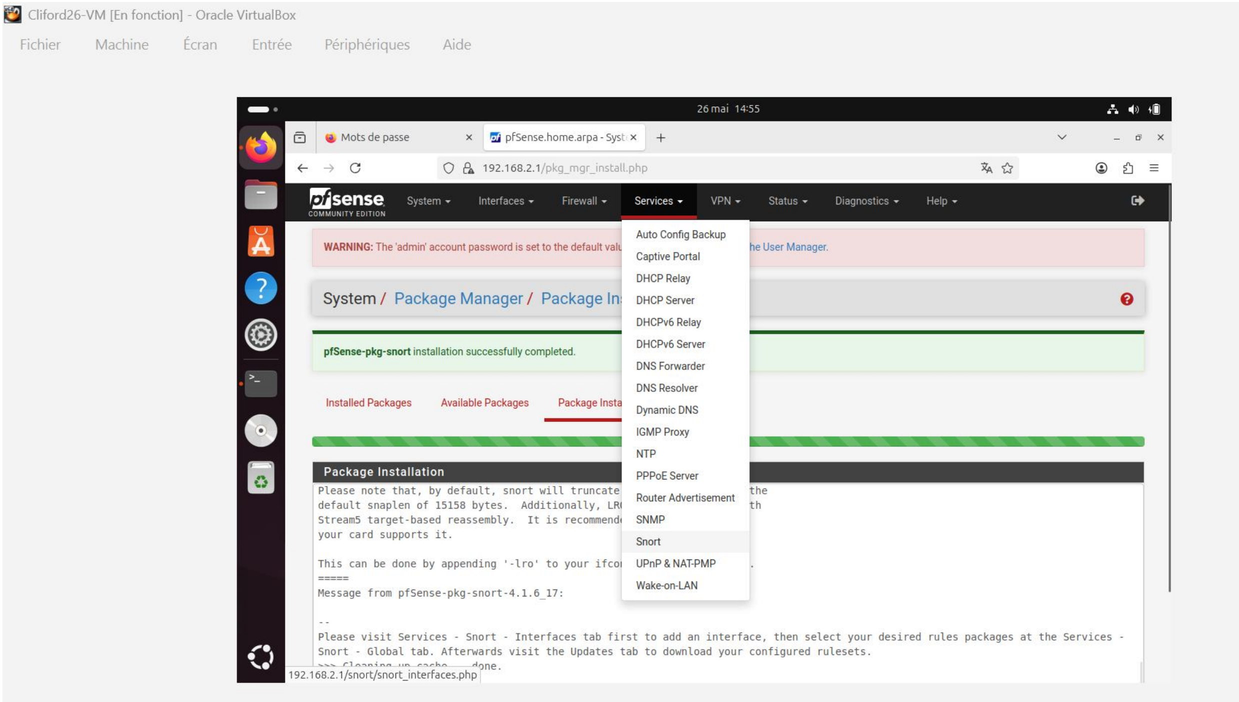
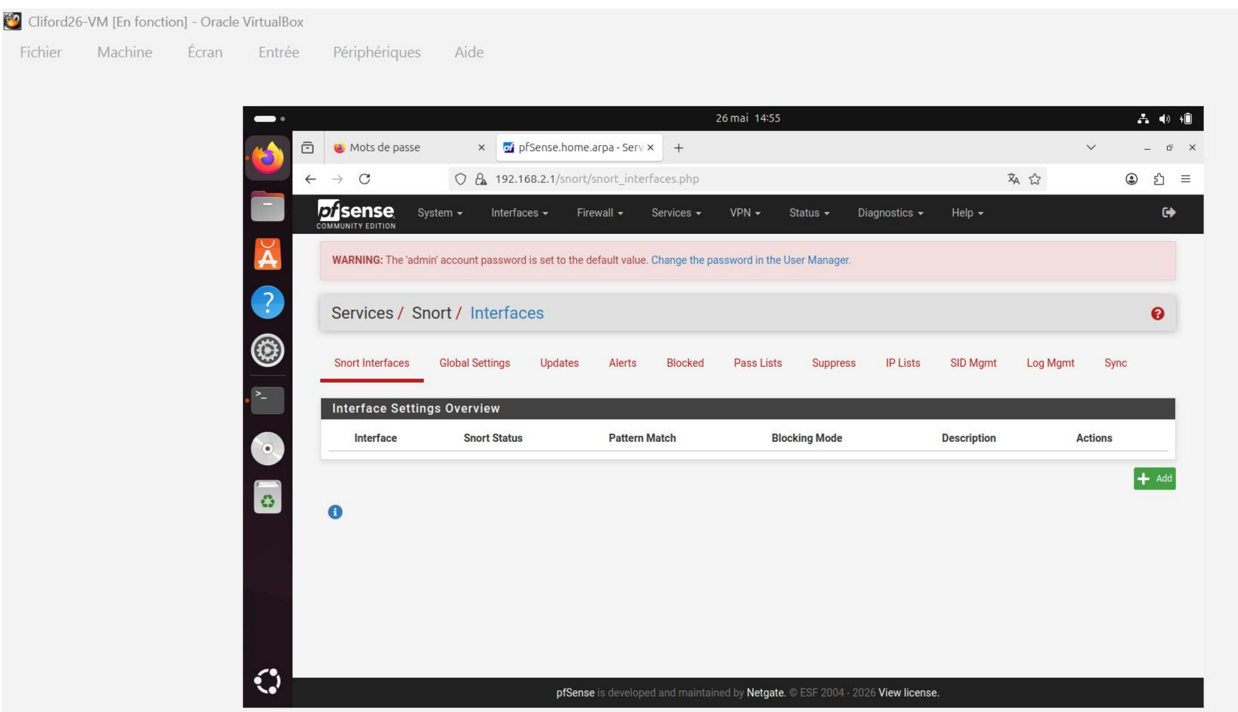
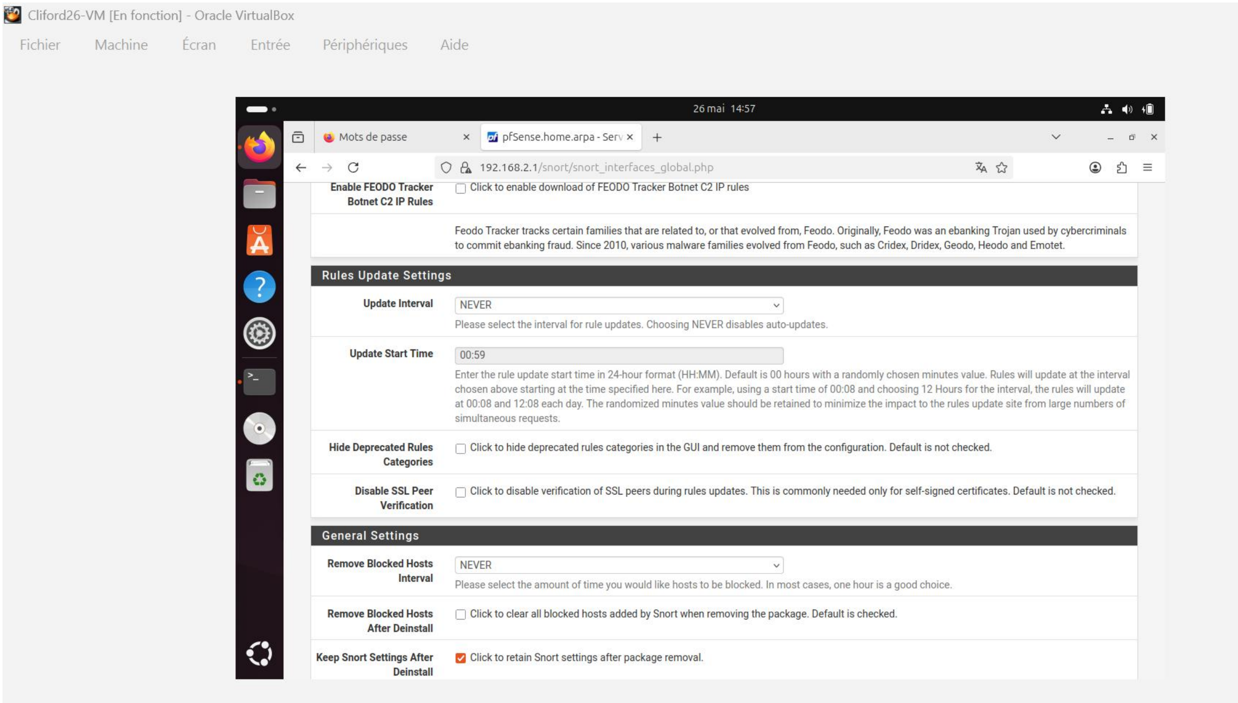
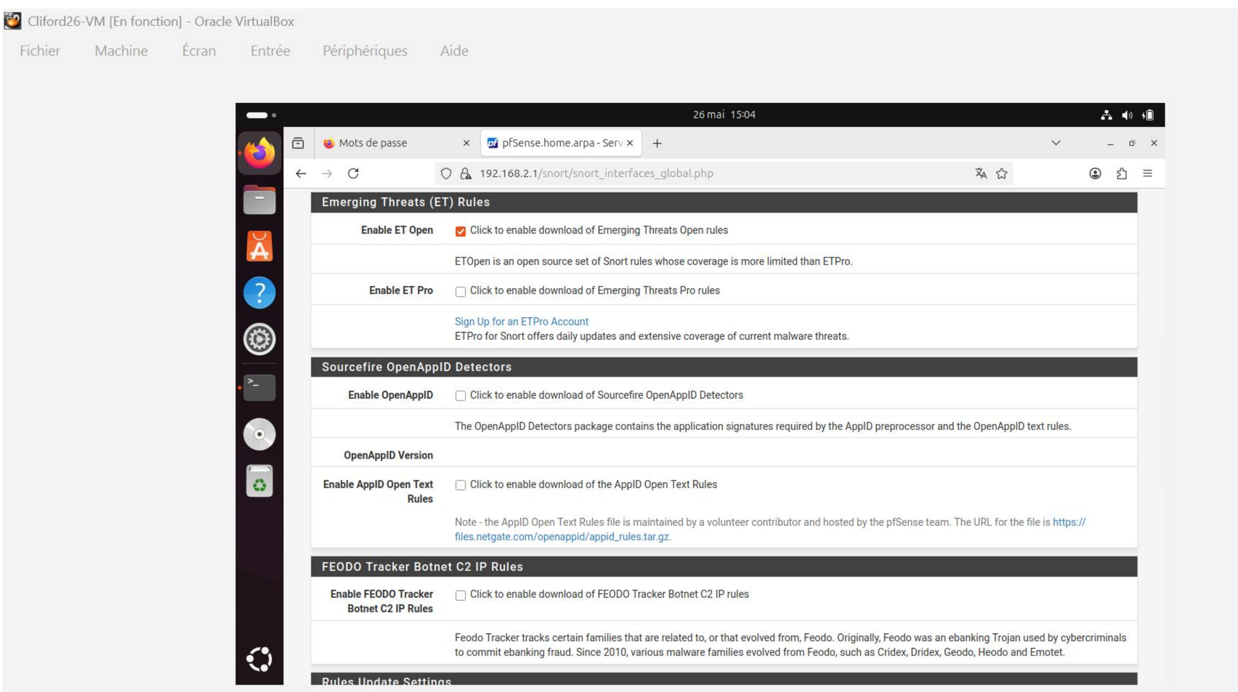
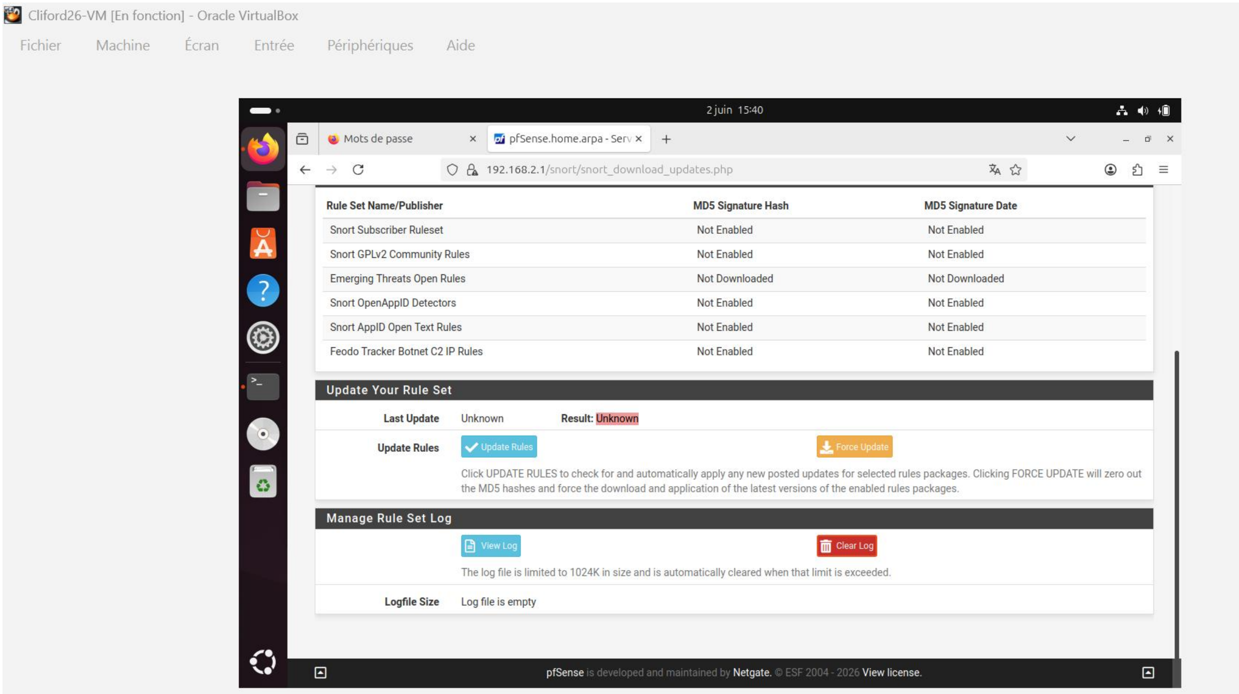


Étape 2 : Configuration des signatures et règles de sécurité




Cliford26-VM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
26 mai 14:57
Mots de passe
pfSense.home.arpa - Serv
192.168.2.1/snort/snort_interfaces_global.php
Enable FEODO Tracker Botnet C2 IP Rules
Click to enable download of FEODO Tracker Botnet C2 IP rules
Feodo Tracker tracks certain families that are related to, or that evolved from, Feodo. Originally, Feodo was an ebanking Trojan used by cybercriminals to commit ebanking fraud. Since 2010, various malware families evolved from Feodo, such as Cridex, Dridex, Geodo, Heodo and Emotet.
Rules Update Settings
Update Interval
NEVER
Please select the interval for rule updates. Choosing NEVER disables auto-updates.
Update Start Time
00:59
Enter the rule update start time in 24-hour format (HH:MM). Default is 00 hours with a randomly chosen minutes value. Rules will update at the interval chosen above starting at the time specified here. For example, using a start time of 00:08 and choosing 12 Hours for the interval, the rules will update at 00:08 and 12:08 each day. The randomized minutes value should be retained to minimize the impact to the rules update site from large numbers of simultaneous requests.
Hide Deprecated Rules Categories
Click to hide deprecated rules categories in the GUI and remove them from the configuration. Default is not checked.
Disable SSL Peer Verification
Click to disable verification of SSL peers during rules updates. This is commonly needed only for self-signed certificates. Default is not checked.
General Settings
Remove Blocked Hosts Interval
NEVER
Please select the amount of time you would like hosts to be blocked. In most cases, one hour is a good choice.
Remove Blocked Hosts After Deinstall
Click to clear all blocked hosts added by Snort when removing the package. Default is checked.
Keep Snort Settings After Deinstall
Click to retain Snort settings after package removal.


Cliford26-VM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
26 mai 15:04
Mots de passe
pfSense.home.arpa - Serv
192.168.2.1/snort/snort_interfaces_global.php
Emerging Threats (ET) Rules
Enable ET Open
Click to enable download of Emerging Threats Open rules
ETOpen is an open source set of Snort rules whose coverage is more limited than ETPro.
Enable ET Pro
Click to enable download of Emerging Threats Pro rules
Sign Up for an ETPro Account
ETPro for Snort offers daily updates and extensive coverage of current malware threats.
Sourcefire OpenAppID Detectors
Enable OpenAppID
Click to enable download of Sourcefire OpenAppID Detectors
The OpenAppID Detectors package contains the application signatures required by the AppID preprocessor and the OpenAppID text rules.
OpenAppID Version
Enable AppID Open Text Rules
Click to enable download of the AppID Open Text Rules
Note - the AppID Open Text Rules file is maintained by a volunteer contributor and hosted by the pfSense team. The URL for the file is https://files.netgate.com/openappid/appid_rules.tar.gz.
FEODO Tracker Botnet C2 IP Rules
Enable FEODO Tracker Botnet C2 IP Rules
Click to enable download of FEODO Tracker Botnet C2 IP rules
Feodo Tracker tracks certain families that are related to, or that evolved from, Feodo. Originally, Feodo was an ebanking Trojan used by cybercriminals to commit ebanking fraud. Since 2010, various malware families evolved from Feodo, such as Cridex, Dridex, Geodo, Heodo and Emotet.


Cliford26-VM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
2 juin 15:40
Mots de passe
pfSense.home.arpa - Serv
192.168.2.1/snort/snort_download_updates.php
Rule Set Name/Publisher | MD5 Signature Hash | MD5 Signature Date
Snort Subscriber Ruleset | Not Enabled | Not Enabled
Snort GPLv2 Community Rules | Not Enabled | Not Enabled
Emerging Threats Open Rules | Not Downloaded | Not Downloaded
Snort OpenAppID Detectors | Not Enabled | Not Enabled
Snort AppID Open Text Rules | Not Enabled | Not Enabled
Feodo Tracker Botnet C2 IP Rules | Not Enabled | Not Enabled
Update Your Rule Set
Last Update Unknown Result: Unknown
Update Rules
Update Rules
Force Update
Click UPDATE RULES to check for and automatically apply any new posted updates for selected rules packages. Clicking FORCE UPDATE will zero out the MD5 hashes and force the download and application of the latest versions of the enabled rules packages.
Manage Rule Set Log
View Log
Clear Log
The log file is limited to 1024K in size and is automatically cleared when that limit is exceeded.
Logfile Size Log file is empty
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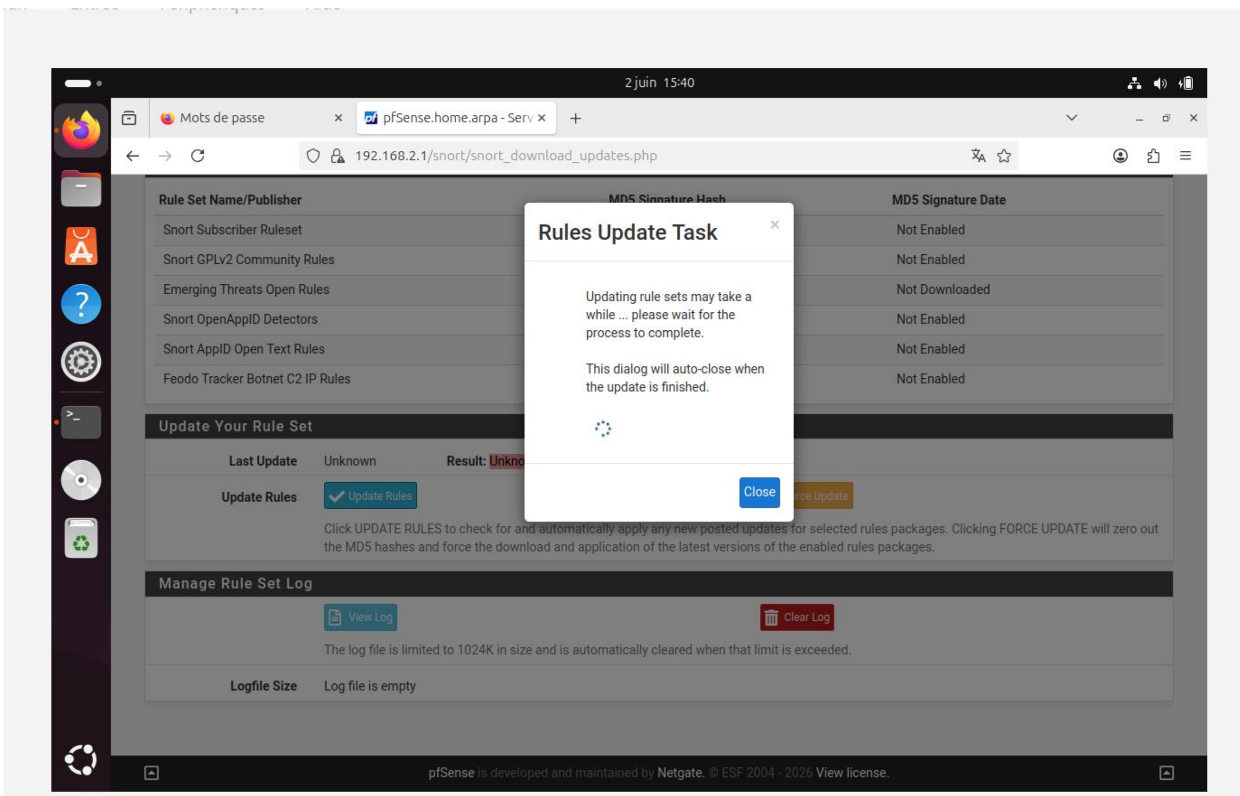
2 juin 15:40
Mots de passe
pfSense.home.arpa - Serv
192.168.2.1/snort/snort_download_updates.php
Rule Set Name/Publisher
MD5 Signature Date
Snort Subscriber Ruleset
Snort GPLv2 Community Rules
Emerging Threats Open Rules
Snort OpenAppID Detectors
Snort AppID Open Text Rules
Feodo Tracker Botnet C2 IP Rules
Not Enabled
Not Enabled
Not Downloaded
Not Enabled
Not Enabled
Not Enabled
Rules Update Task
Updating rule sets may take a while ... please wait for the process to complete.
This dialog will auto-close when the update is finished.
Close
Update Your Rule Set
Last Update Unknown
Update Rules
Update Rules
Click UPDATE RULES to check for and automatically apply any new posted updates for selected rules packages. Clicking FORCE UPDATE will zero out the MD5 hashes and force the download and application of the latest versions of the enabled rules packages.
Manage Rule Set Log
View Log
Clear Log
The log file is limited to 1024K in size and is automatically cleared when that limit is exceeded.
Logfile Size Log file is empty
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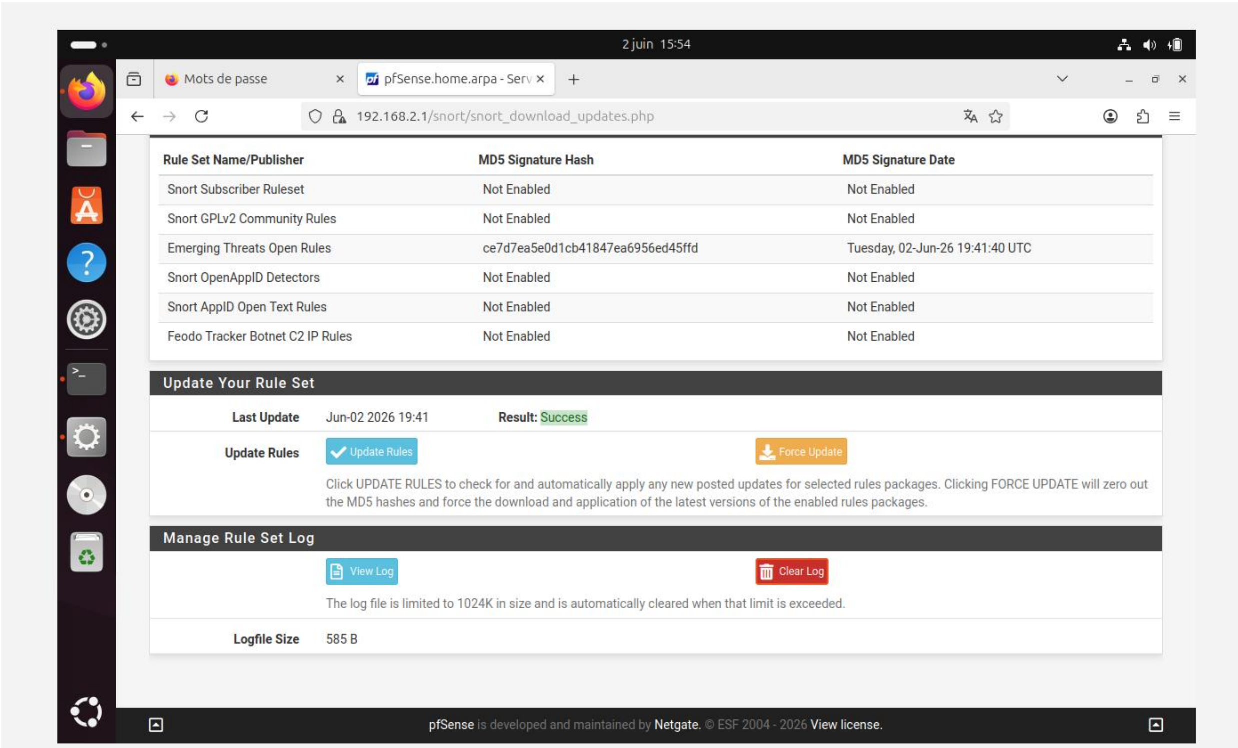
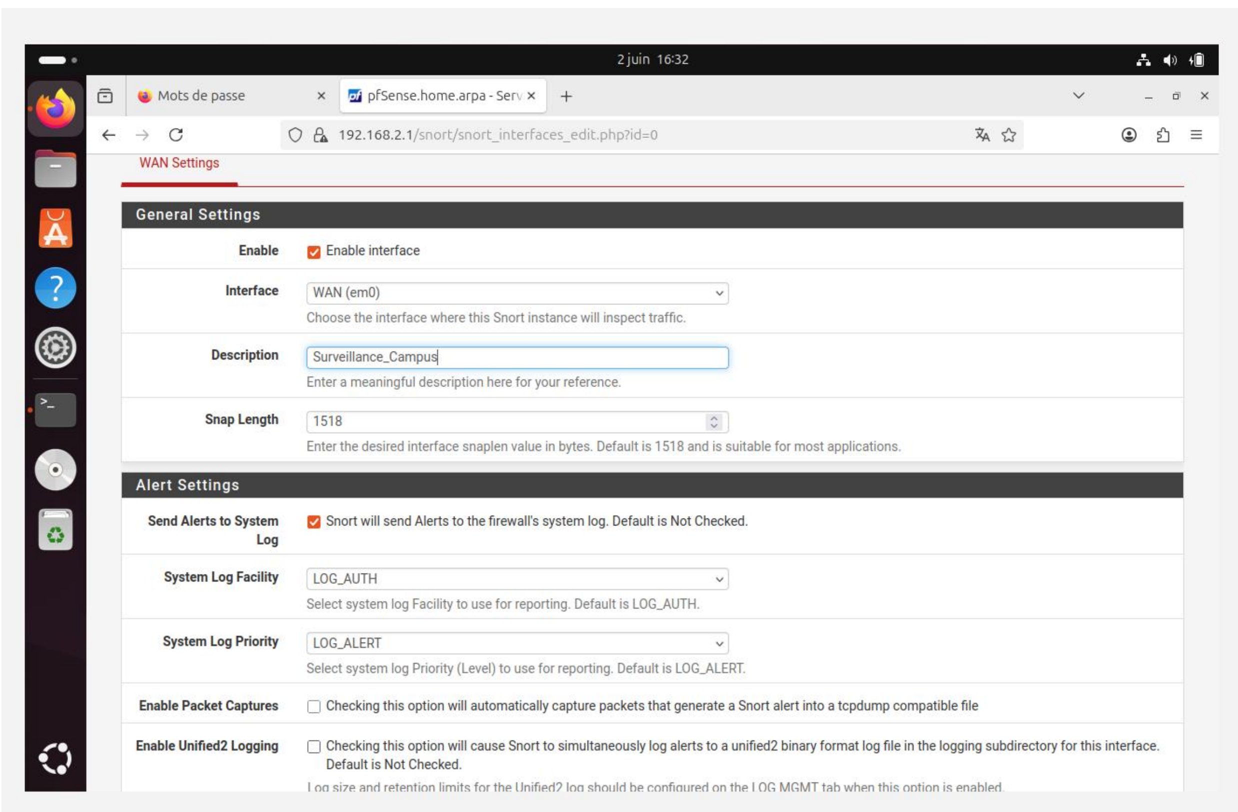
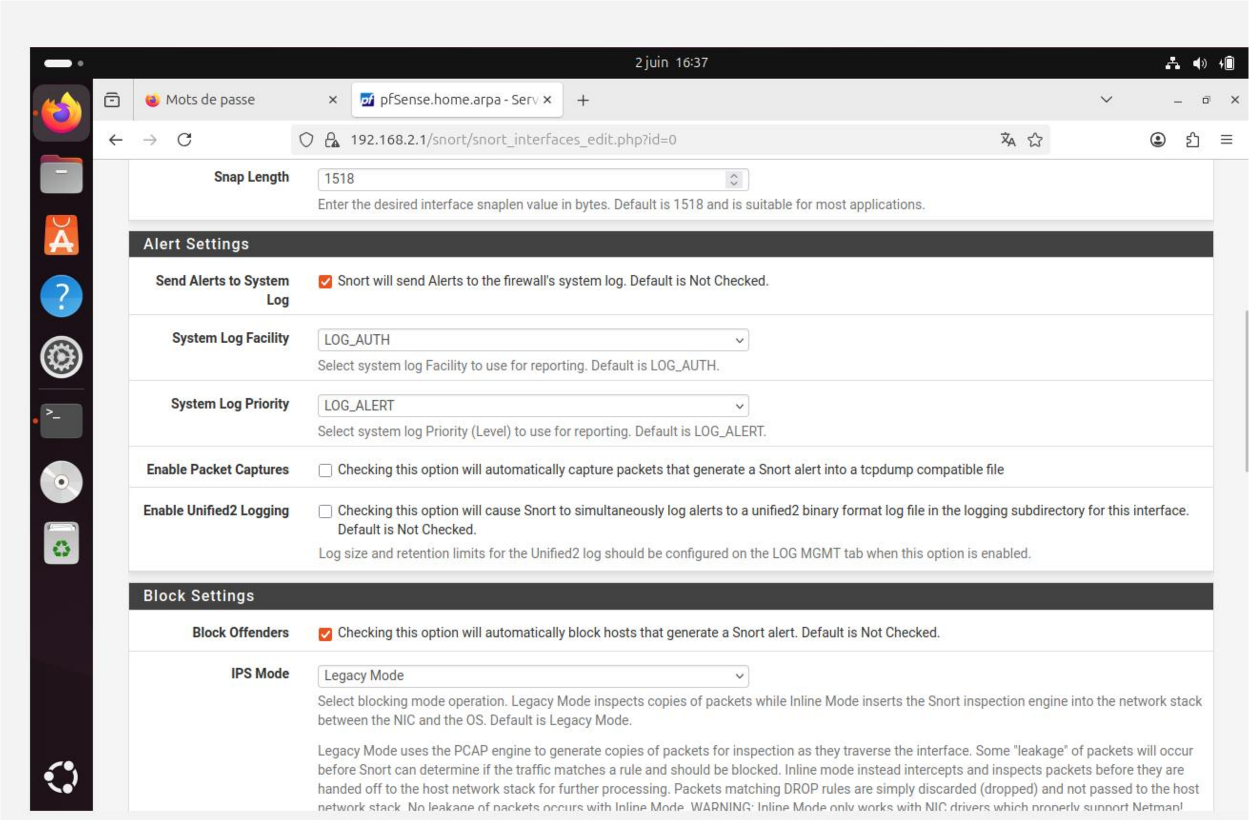
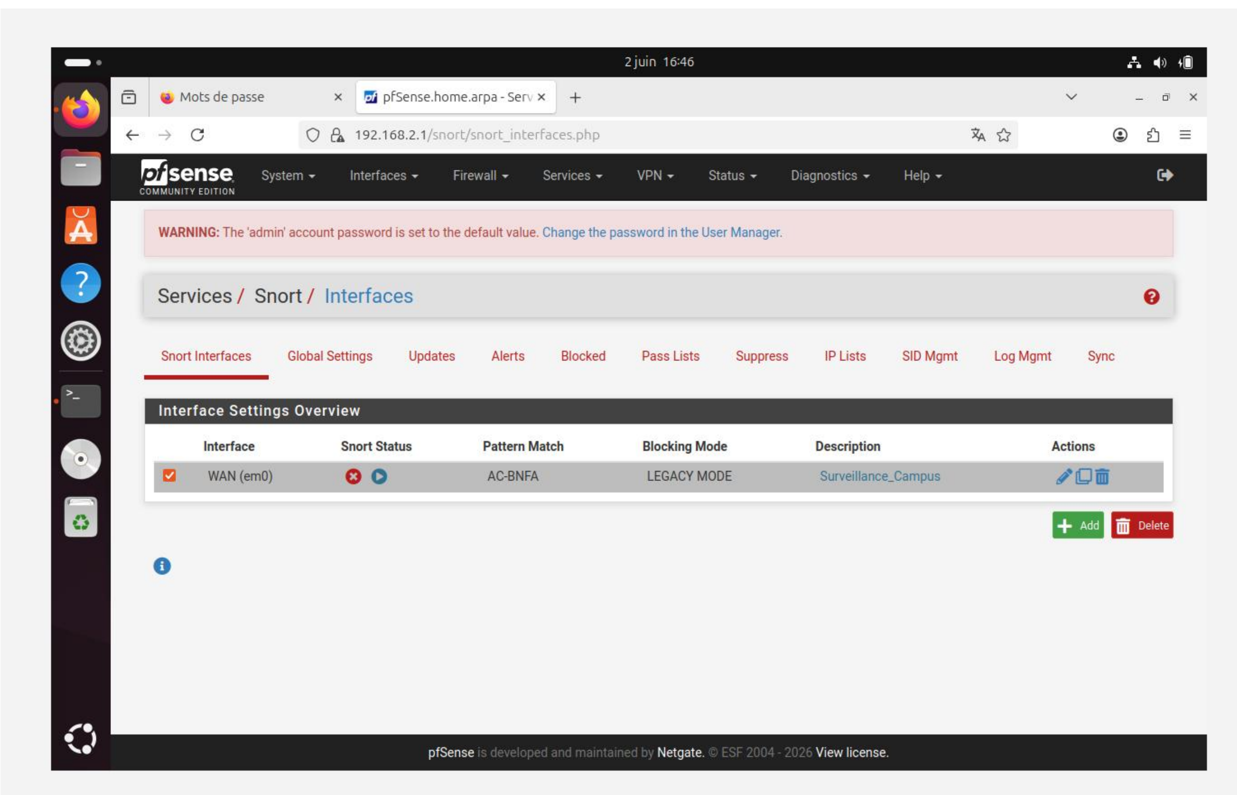


Étape 3 : Activation de Snort sur les interfaces réseau

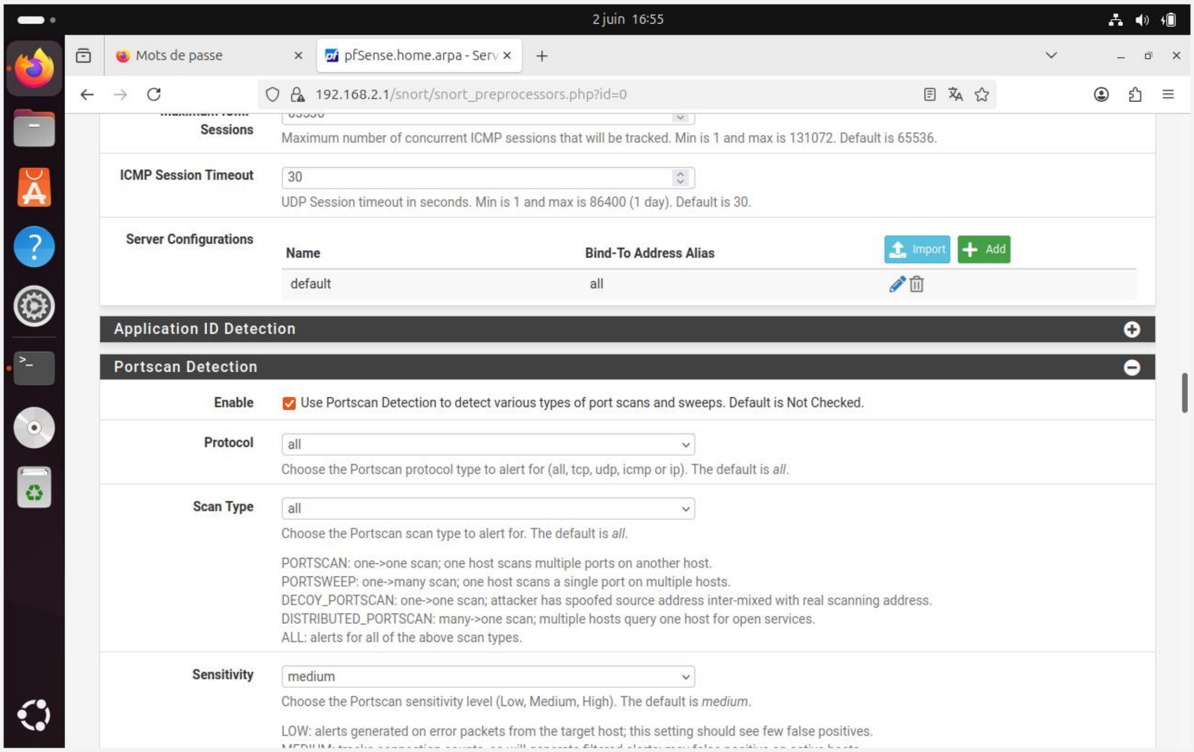


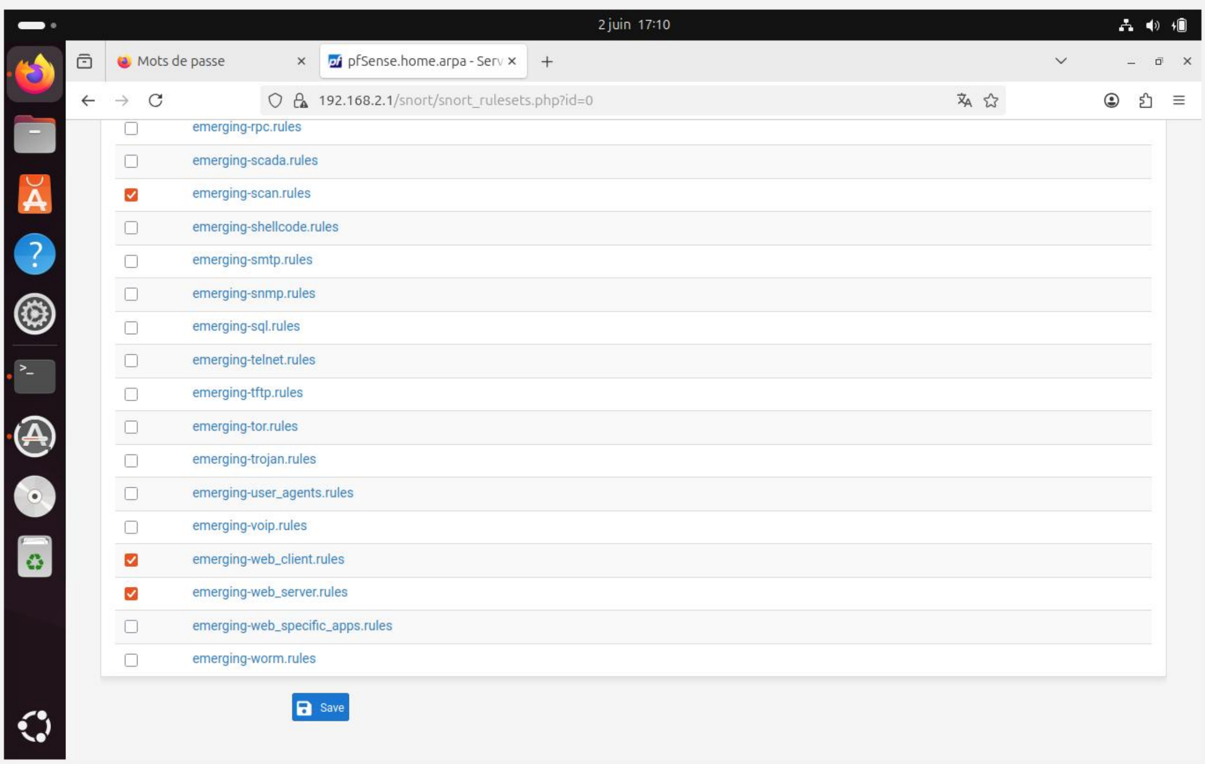
Étape 4 : Passer du mode Détection (IDS) au mode Blocage (IPS)



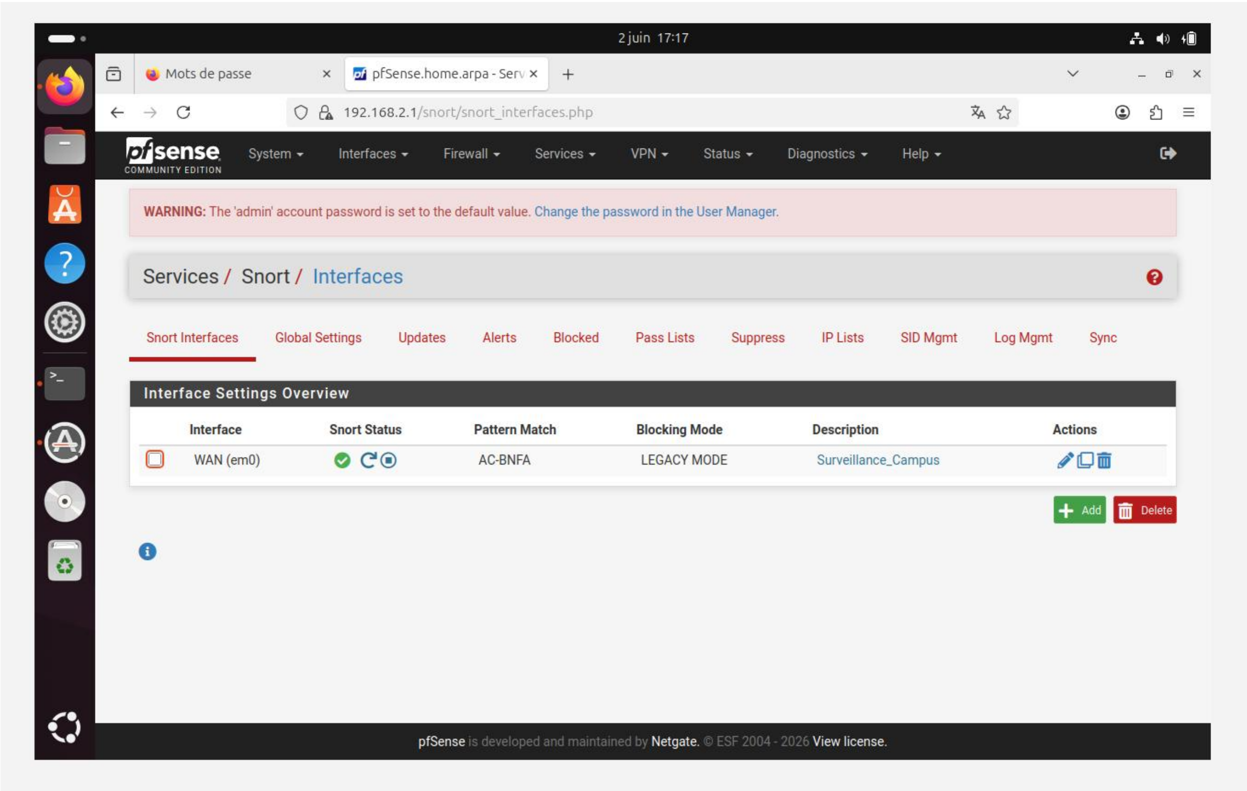


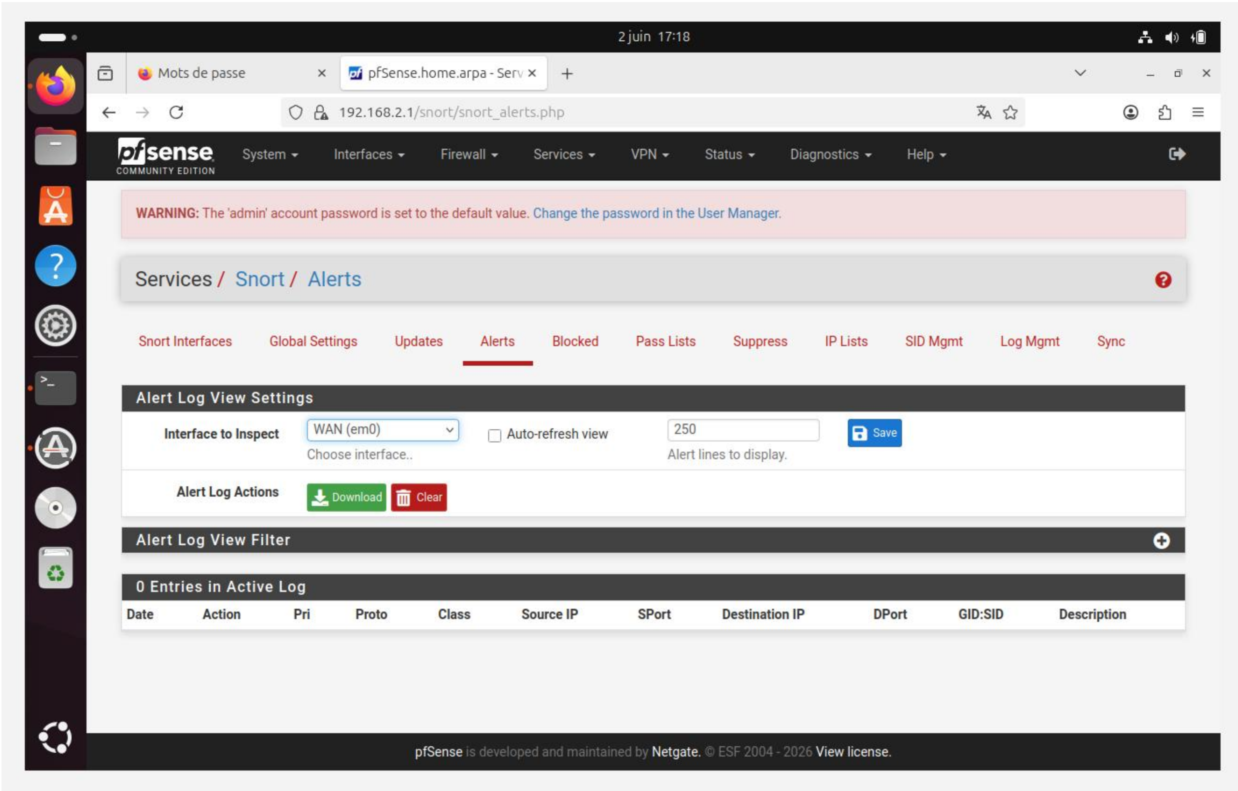
Étape 5 : Réponse ciblée aux menaces spécifiques de l'IUS





Étape 6 : Démarrage et vérification




2 juin 17:18
Mots de passe
pfSense.home.arpa - Serv
192.168.2.1/snort/snort_alerts.php
System Interfaces Firewall Services VPN Status Diagnostics Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Services / Snort / Alerts
Snort Interfaces Global Settings Updates Alerts Blocked Pass Lists Suppress IP Lists SID Mgmt Log Mgmt Sync
Alert Log View Settings
Interface to Inspect
WAN (em0)
Choose interface..
Auto-refresh view
250
Alert lines to display.
Save
Alert Log Actions
Download
Clear
Alert Log View Filter
0 Entries in Active Log
Date Action Pri Proto Class Source IP SPort Destination IP DPort GID:SID Description
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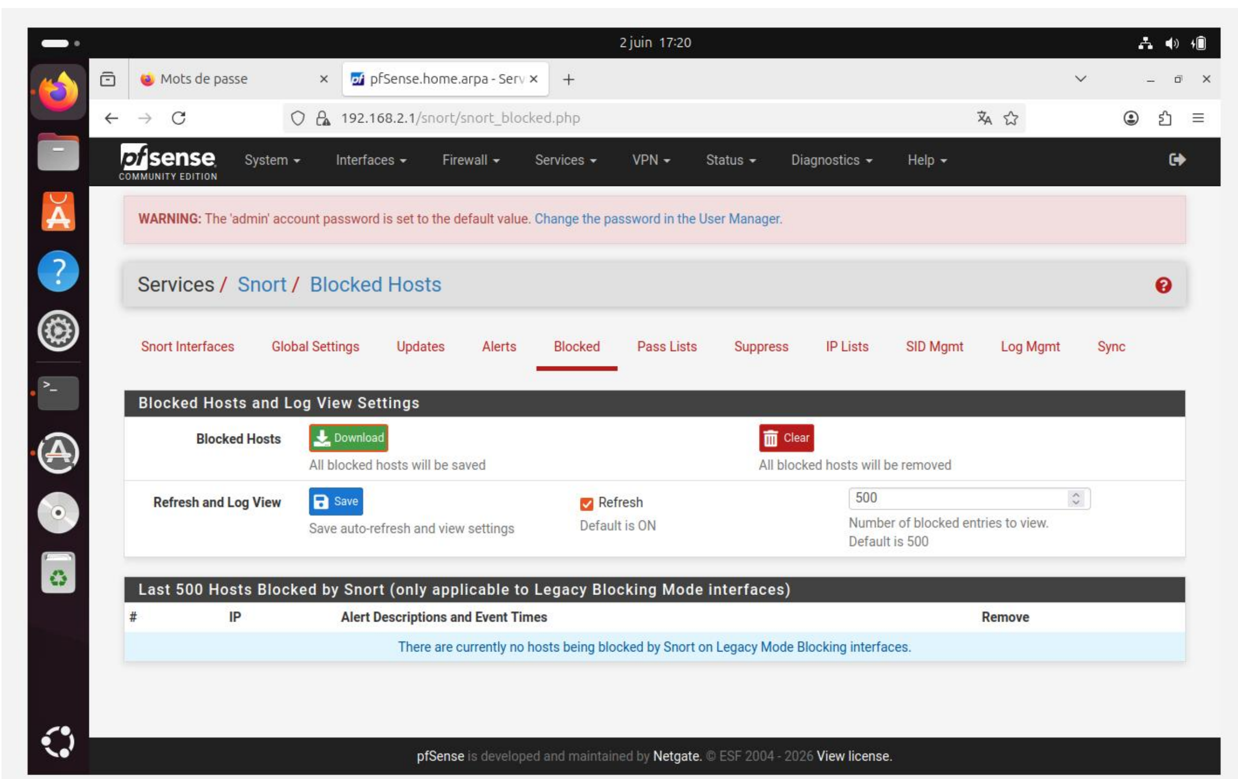
2 juin 17:20
Mots de passe
pfSense.home.arpa - Serv
192.168.2.1/snort/snort_blocked.php
System Interfaces Firewall Services VPN Status Diagnostics Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Services / Snort / Blocked Hosts
Snort Interfaces Global Settings Updates Alerts Blocked Pass Lists Suppress IP Lists SID Mgmt Log Mgmt Sync
Blocked Hosts and Log View Settings
Blocked Hosts
Download
All blocked hosts will be saved
Clear
All blocked hosts will be removed
Refresh and Log View
Save
Save auto-refresh and view settings
Refresh
Default is ON
500
Number of blocked entries to view.
Default is 500
Last 500 Hosts Blocked by Snort (only applicable to Legacy Blocking Mode interfaces)
IP Alert Descriptions and Event Times Remove
There are currently no hosts being blocked by Snort on Legacy Mode Blocking interfaces.
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Questions? Reponse !

1. Différence entre IDS et IPS ?

- **IDS (Intrusion Detection System - Système de Détection d'Intrusions) :** Il agit comme une caméra de surveillance. Il écoute le trafic réseau de manière passive, détecte les activités suspectes et génère une **alerte**, mais il ne bloque pas l'attaque par lui-même.
- **IPS (Intrusion Prevention System - Système de Prévention d'Intrusions) :** Il agit comme un agent de sécurité. Placé en coupure sur le réseau (ou configuré pour interagir activement), il analyse le trafic et prend immédiatement des mesures pour **bloquer ou rejeter la menace** (par exemple en bannissant l'adresse IP de l'attaquant) dès qu'elle est détectée.

2. Pourquoi installer Snort sur WAN ?

Installer Snort sur l'interface **WAN** (l'interface connectée à Internet) permet de :

- **Sécuriser la frontière extérieure :** Intercepter, analyser et bloquer les attaques (scans de ports, requêtes malveillantes) *avant* qu'elles ne parviennent à pénétrer à l'intérieur du réseau local de l'université.
- **Économiser les ressources internes :** En stoppant le trafic malveillant dès la porte d'entrée, on évite que les serveurs internes ou les pare-feux secondaires ne perdent du temps à traiter des paquets inutiles ou dangereux.

3. Qu’est-ce qu’une attaque ICMP flood ?

Une attaque **ICMP flood** (ou submersion par requêtes ICMP) est une forme d'attaque par déni de service (**DoS**).

- L'attaquant envoie une quantité massive et ininterrompue de paquets de demande d'écho (`Ping`) à une machine ou une passerelle du réseau.
- Le système ciblé s'épuise en essayant de répondre à chaque demande (paquets *Echo Reply*), ce qui sature sa bande passante et ses ressources CPU, rendant le réseau totalement indisponible ou extrêmement lent pour les utilisateurs légitimes.

4. Comment Snort détecte une intrusion ?

Snort utilise principalement la **détection par signatures**. Il fonctionne de la manière suivante :

- Il inspecte le contenu et les en-têtes de chaque paquet réseau qui traverse l'interface.
- Il compare ces données en temps réel avec une base de données de règles prédéfinies (comme les règles *Emerging Threats* ou *Snort VRT*).

- Si la structure d'un paquet correspond exactement à la description d'une menace connue dans la base (la "signature"), Snort déclenche instantanément l'action associée (Alerte ou Blocage).

5. Que faire en cas de faux positif ?

Un faux positif se produit lorsque Snort bloque ou alerte sur un trafic légitime (par exemple, une mise à jour logicielle saine prise à tort pour un virus). Pour corriger cela, on peut :

- **Utiliser la liste blanche (Pass Lists) :** Ajouter l'adresse IP légitime dans une *Pass List* pour que Snort ignore ses flux et ne la bloque jamais.
- **Désactiver la règle spécifique :** Repérer l'identifiant de la règle (**SID**) qui a causé l'erreur dans l'onglet *Alerts*, et la désactiver spécifiquement pour l'interface concernée.
- **Modifier ou filtrer la règle (Suppress) :** Utiliser l'onglet **Suppress** pour dire à Snort de ne plus déclencher d'alerte pour cette règle précise lorsqu'elle provient de ou se dirige vers une adresse IP particulière.